Topological and Metric Valuation Theory on Regular Tessellations: From Gray's Cell Groups to N-Dimensional Dual Cell Valuations

A Formal Treatise on Digital Topology, Valuation Theory, and Local Computability

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Abstract

In his 1971 foundational paper *Local Properties of Binary Images in Two Dimensions*, Stephen B. Gray established that fundamental metric and topological properties of two-dimensional continuous and discrete binary images—including area, perimeter, and the topological Euler characteristic—are strictly *locally countable*. Gray based his discrete framework on *cell groups*, which are dual polygons centered at grid intersections connecting adjacent bit cells. This monograph presents a unified, mathematically rigorous formalization and higher-dimensional generalization of Gray's approach. We demonstrate that cell groups are the planar instantiation of *dual Voronoi cell decompositions* on cellular complexes.

Within this unified valuation-theoretic framework, we formalize:

1. Continuous binary topology, Jordan ray parity, continuous image triangulations, and the additive Hadwiger valuation properties for the Euler characteristic χ, area A, and perimeter P, alongside sweep-line Morse–Bott critical point index theorems ($\chi = n(X_\theta) - n(V_\theta)$).
2. The geometry of dual cell groups on the three regular planar tessellations $(6,3)$, $(4,4)$, and $(3,6)$, proving that hexagonal cell groups are universally linearly separable (0% inseparability), whereas square grids exhibit diagonal ambiguities (12.5% inseparability) and triangular grids exhibit 50% inseparability.
3. Exact discrete valuation formulas for χ, A, and P, including Cauchy–Crofton continuous perimeter approximations ($P_a = \frac{T_1+T_2}{1.24}$) and corner-cutting smooth square grid metrics.
4. Topological and metric differentials ($E^*, W^*, Z^*, H^*, A^*, P^*$), simple point characterizations, and subfield decomposition theorems for topology-preserving parallel skeletonization.
5. Geometric shape descriptors for elongated and fibrous patterns (line length $k = P/2$, width $w = 2A/P$, aspect ratio $k/w = P^2/(4A)$, and isoperimetric circularity $\mathcal{D} = \frac{P^2}{4\pi A} \geq 1$).
6. Generalization to arbitrary N-dimensional uniform cellular complexes and hypercubic lattices $\mathbb{Z}^N$ via dual Voronoi cell valuations and Forman's discrete Morse spherical link formula ($\Delta\chi(Q) = 1 - \chi(\mathrm{Lk}(v, K))$), establishing the universal denominator 2^N.
7. Complete 3-dimensional volume processing on $\mathbb{Z}^3$, including the classification of all 256 bit octocubes under octahedral symmetry O_h into 22 equivalence classes, 3D surface area valuations, and 3D digital connectivity pairs.
8. The theory of local computability via the Local Property Counter (LPC) complexity 4-tuple (D, G, R, M), proving universal diameter-2 sufficiency across all dimensions, with comparisons to Minsky–Papert perceptron order limitations.

Contents

1 Introduction and Mathematical Foundations **3**
- 1.1 The Problem of Local Computability in Digital Geometry . . . 3
- 1.2 The Cell-Group Concept and the Dual Valuation Paradigm . . . 3
- 1.3 Objectives and Organization of this Monograph . . . 3

2 Continuous Binary Images, Valuations, and Morse Sweeps **4**
- 2.1 Border Curves, Parity, and Binary Regions . . . 4
- 2.2 Continuous Triangulation and the Euler–Poincaré Invariant . . . 5
- 2.3 The Additive Valuation Property and Non-Linear Union Interactions . . . 5
- 2.4 Morse–Bott Critical Point Sweeps . . . 6

3 Planar Regular Tessellations and the Geometry of Cell Groups **7**
- 3.1 Classification of Regular Planar Tessellations (p, q) . . . 7
- 3.2 Formal Definition of Dual Cell Groups . . . 7
- 3.3 Linear Separability of Cell Groups and Connectivity Ambiguities . . . 7

4 Discrete Local Valuations on Two-Dimensional Regular Lattices **8**
- 4.1 The 2D Dual Decomposition Valuation Principle . . . 8
- 4.2 The Hexagonal Lattice $(6, 3)$ . . . 8
  - 4.2.1 Multiplicity Parameters . . . 9
- 4.3 The Square Lattice $(4, 4)$ . . . 9
  - 4.3.1 Multiplicity Parameters . . . 10
- 4.4 The Triangular Lattice $(3, 6)$ . . . 11

5 Local Differentials and Parallel Skeletonization **11**
- 5.1 Definition and Locality of Differentials . . . 11
- 5.2 Hexagonal Differentials . . . 11
- 5.3 Square Lattice Differentials . . . 12
- 5.4 Subfield Decomposition and Parallel Skeletonization . . . 12

6 Geometric Shape Characterization via Locally Countable Properties **12**

7 N-Dimensional Dual Voronoi Valuation Framework **13**
- 7.1 Geometric and Topological Setup . . . 13
- 7.2 The Dual Voronoi Partition . . . 14
- 7.3 The Universal N-Dimensional Local Valuation Theorem . . . 14

8 Specialization to 3-Dimensional Volume Processing on $\mathbb{Z}^3$ **15**
- 8.1 Combinatorial Parameters and Denominator Clearing . . . 15
- 8.2 Complete Classification of the 22 Octocube Equivalence Classes . . . 15
- 8.3 3D Surface Area Valuation . . . 15

9 Theory of Local Computability and Local Property Counters (LPC) **16**
- 9.1 The Local Property Counter (LPC) Formalism . . . 16
- 9.2 Fundamental Complexity Theorems for Local Counting . . . 16
- 9.3 Master Classification Table of Local Invariants Across Dimensions . . . 17

10 Hardware Architecture of the Binary Image Processor (BIP) **17**

11 Conclusion **18**

1 Introduction and Mathematical Foundations

1.1 The Problem of Local Computability in Digital Geometry

A central question in computational geometry, pattern recognition, and discrete topology is: *Which global geometric and topological properties of a spatial pattern can be computed by accumulating purely local, finite-window measurements?*

In 1969, Marvin Minsky and Seymour Papert published their influential work *Perceptrons* [2], proving that the fundamental topological predicate of **connectedness** is not of finite order. That is, no translation-invariant linear threshold device with finite diameter or finite input order can determine whether an arbitrary digitized binary image consists of a single connected component.

Two years later, in 1971, Stephen B. Gray published *Local Properties of Binary Images in Two Dimensions* in the *IEEE Transactions on Computers* [1]. Gray made the crucial observation that while deciding connectedness globally requires unbounded spatial support, the **Euler characteristic** (or Euler number):

$$\chi = B - L \tag{1}$$

—defined in the Euclidean plane as the number of connected components (objects B) minus the number of enclosed cavities (holes L)—is strictly **locally countable**. That is, the global integer invariant χ can be evaluated by sweeping a small, finite-diameter neighborhood window across the image, evaluating a local valuation on each configuration, and summing the results.

Furthermore, Gray demonstrated that key metric invariants, such as total area A and boundary perimeter P, are also locally countable. By combining these local measurements, one can compute scale-invariant and affine-invariant shape descriptors (such as average object area, average perimeter, fiber width, and aspect ratio) during a single sequential scan through memory.

1.2 The Cell-Group Concept and the Dual Valuation Paradigm

To derive local counting formulas on regular digital lattices, Gray introduced the geometric concept of **cell groups**. In Gray's terminology:

- A **bit cell** is a primal polygon of the tessellation (e.g., a square or a regular hexagon) carrying a binary state 0 or 1.
- A **cell group** is a dual polygon formed by connecting the geometric centers of all adjacent bit cells meeting at a common grid vertex.

By analyzing how primal vertices, edges, and polygonal faces are shared across adjacent cell groups, Gray established explicit combinatorial formulas for computing χ, A, and P on the hexagonal $(6, 3)$ and square $(4, 4)$ lattices. He addressed the diagonal connectivity ambiguities inherent in the square grid, defined the **Euler differential** $E^*(U) = \Delta E/\Delta U$ to govern topology-preserving parallel skeletonization, and characterized hardware measurement complexity through the **Local Property Counter** (LPC) formalism.

1.3 Objectives and Organization of this Monograph

The purpose of this monograph is to provide a complete, unified, and mathematically rigorous treatment of Gray's 1971 paper and to generalize its principles to arbitrary N-dimensional cellular complexes and lattices.

The structure of this work is as follows:

- Section 2 establishes the continuous topology of binary planar images, continuous image triangulations, the Hadwiger valuation property, and Morse–Bott critical point sweep theorems.

- Section 3 formalizes the geometry of the three regular planar tessellations (p, q), the dual cell-group construction, and the linear separability of cell groups.
- Section 4 derives the explicit local valuation formulas for χ, A, and P on the hexagonal $(6, 3)$, square $(4, 4)$, and triangular $(3, 6)$ lattices.
- Section 5 formalizes the theory of local differentials $(E^*, W^*, Z^*, H^*, A^*, P^*)$ and proves the subfield decomposition theorem for collision-free parallel thinning.
- Section 6 develops geometric shape descriptors for fibrous and elongated patterns.
- Section 7 generalizes the cell-group valuation theory to arbitrary N-dimensional uniform complexes and hypercubic lattices $\mathbb{Z}^N$, proving the Universal Local Valuation Theorem and connecting to Forman's discrete Morse theory.
- Section 8 specializes the theory to 3D cubic lattices $\mathbb{Z}^3$, providing the complete classification of all 256 octocube configurations into 22 O_h equivalence classes.
- Section 9 establishes the theory of local computability, proving universal diameter-2 sufficiency and comparing order bounds.
- Section 10 reviews the hardware architecture of the Binary Image Processor (BIP).

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2 Continuous Binary Images, Valuations, and Morse Sweeps

We first formalize the topological and measure-theoretic foundations of continuous binary spatial patterns in $\mathbb{R}^2$.

2.1 Border Curves, Parity, and Binary Regions

Definition 2.1 (Border Curve and Border Set)**.** A *border curve* $\gamma \subset \mathbb{R}^2$ is the image of a simple closed Jordan curve, parameterized by an injective embedding $f : S^1 \hookrightarrow \mathbb{R}^2$. A *border set* $\mathcal{B} = \{\gamma_1, \ldots, \gamma_m\}$ is a finite set of pairwise disjoint border curves:

$$\gamma_i \cap \gamma_j = \emptyset \quad \forall i \neq j. \tag{2}$$

By the Jordan Curve Theorem, each curve γ_i partitions $\mathbb{R}^2 \setminus \gamma_i$ into two connected open domains: a bounded interior $\mathrm{Int}(\gamma_i)$ and an unbounded exterior $\mathrm{Ext}(\gamma_i)$.

Definition 2.2 (Jordan Ray Parity and Continuous Binary Image)**.** Let $\mathcal{B} = \{\gamma_1, \ldots, \gamma_m\}$ be a border set. For any point $x \in \mathbb{R}^2 \setminus \bigcup_{i=1}^m \gamma_i$, let $\rho_x : [0, \infty) \to \mathbb{R}^2$ be any smooth ray starting at $\rho_x(0) = x$ and escaping to infinity such that ρ_x intersects all curves γ_i transversally. The *parity* $P(x) \in \{0, 1\}$ is:

$$P(x) \coloneqq \left| \rho_x([0, \infty)) \cap \bigcup_{i=1}^{m} \gamma_i \right| \pmod 2. \tag{3}$$

The induced *continuous binary image* $I : \mathbb{R}^2 \to \{0, 1\}$ is defined by:

$$I(x) \coloneqq \begin{cases} P(x) & \text{if } x \in \mathbb{R}^2 \setminus \bigcup_{i=1}^m \gamma_i, \\ 0 & \text{on boundary curves } \bigcup_{i=1}^m \gamma_i. \end{cases} \tag{4}$$

The *foreground* is $X \coloneqq \{x \in \mathbb{R}^2 \mid I(x) = 1\}$, and the *background* is $X^c \coloneqq \mathbb{R}^2 \setminus X$.

- An *object* is a connected component of X. The number of objects is denoted by $B = b_0(X) = \dim H_0(X;\mathbb{R})$.
- A *hole* is a bounded connected component of X^c. The number of holes is denoted by $L = b_1(X) = \dim H_1(X;\mathbb{R})$.

Proposition 2.3 (Topological Invariance of Ray Parity)**.** *The parity $P(x)$ is independent of the choice of transverse ray ρ_x, and is locally constant on each connected component of $\mathbb{R}^2 \setminus \bigcup_{i=1}^m \gamma_i$.*

Proof. Let ρ_x^1 and ρ_x^2 be two distinct rays from x to infinity. The formal sum $\rho_x^1 - \rho_x^2$ defines a 1-cycle in the one-point compactification $S^2 = \mathbb{R}^2 \cup \{\infty\}$. The modulo-2 intersection number of any 1-cycle with the closed 1-manifold $\bigcup \gamma_i$ on S^2 is homologically trivial. Thus $|\rho_x^1 \cap \bigcup \gamma_i| \equiv |\rho_x^2 \cap \bigcup \gamma_i| \pmod 2$. □

2.2 Continuous Triangulation and the Euler–Poincaré Invariant

Definition 2.4 (Valid Image Triangulation)**.** Let $X \subset \mathbb{R}^2$ be a compact continuous binary foreground. A *triangulation* $\mathcal{K}$ of X is a finite 2-dimensional simplicial complex whose underlying space $|\mathcal{K}|$ is homeomorphic to $\bar{X}$, such that every open 2-simplex $\sigma_2 \in \mathcal{C}_2(\mathcal{K})$ has $I(x) = 1$ for all $x \in \sigma_2$, and the boundary $\partial X = \bigcup \gamma_i$ is a subcomplex of $\mathcal{K}$.

Theorem 2.5 (Euler–Poincaré Characteristic for Binary Images)**.** *Let n_0, n_1, n_2 denote the total number of 0-simplices (vertices), 1-simplices (edges), and 2-simplices (faces) in any valid triangulation $\mathcal{K}$ of X. Then the alternating sum:*

$$E(X) \coloneqq n_0 - n_1 + n_2 \tag{5}$$

is an intrinsic topological invariant equal to the Euler characteristic:

$$E(X) = \chi(X) = b_0(X) - b_1(X) = B - L. \tag{6}$$

Proof. By the classical Euler–Poincaré formula for finite cell complexes:

$$n_0 - n_1 + n_2 = \sum_{k=0}^{2} (-1)^k \dim H_k(X;\mathbb{R}).$$

For any compact 2D planar region embedded in $\mathbb{R}^2$, the homology groups evaluate to $H_0(X;\mathbb{R}) \cong \mathbb{R}^B$, $H_1(X;\mathbb{R}) \cong \mathbb{R}^L$, and $H_2(X;\mathbb{R}) \cong 0$. Therefore, $n_0 - n_1 + n_2 = B - L$. □

2.3 The Additive Valuation Property and Non-Linear Union Interactions

In the sense of Hadwiger's convex geometry [3], a functional $\Phi : \mathcal{P}(\mathbb{R}^2) \to \mathbb{R}$ is an *additive valuation* if for any two valid regions I_1, I_2:

$$\Phi(I_1 \cup I_2) = \Phi(I_1) + \Phi(I_2) - \Phi(I_1 \cap I_2). \tag{7}$$

Theorem 2.6 (Valuation Property of Euler Characteristic, Area, and Perimeter)**.** *Let $I_1, I_2 \subset \mathbb{R}^2$ be two compact, finitely triangulable binary foregrounds. Then:*

$$E(I_1 \cup I_2) = E(I_1) + E(I_2) - E(I_1 \cap I_2), \tag{8}$$
$$A(I_1 \cup I_2) = A(I_1) + A(I_2) - A(I_1 \cap I_2), \tag{9}$$
$$P(I_1 \cup I_2) = P(I_1) + P(I_2) - P(I_1 \cap I_2), \tag{10}$$

where $A(\cdot)$ is 2D Lebesgue area, and $P(\cdot)$ is 1D boundary Hausdorff length (with interior contact lines counted with multiplicity 2 in the intersection).

Proof. Let $\mathcal{K}$ be a simultaneous triangulation of $I_1 \cup I_2, I_1, I_2$, and $I_1 \cap I_2$. For each simplex dimension $k \in \{0, 1, 2\}$, the indicator functions satisfy:

$$n_k(I_1 \cup I_2) = n_k(I_1) + n_k(I_2) - n_k(I_1 \cap I_2).$$

Taking the alternating sum $\sum_{k=0}^{2}(-1)^k n_k$ yields (8). Equations (9) and (10) follow directly from the additivity of Lebesgue integration: $A(I) = \int_{\mathbb{R}^2} \mathbf{1}_I(x)\,dx$ and $P(I) = \int_{\partial I} ds$. □

Corollary 2.7 (Inclusion-Exclusion Principle for N Images)**.** *For any finite collection of binary images $\{I_1, I_2, \ldots, I_N\}$:*

$$E\left(\bigcup_{i=1}^{N} I_i\right) = \sum_{k=1}^{N}(-1)^{k-1} \sum_{1 \le i_1 < i_2 < \cdots < i_k \le N} E\left(I_{i_1} \cap I_{i_2} \cap \cdots \cap I_{i_k}\right). \tag{11}$$

Lemma 2.8 (Non-Linearity Condition for Multiple Image Additions)**.** *Let I be a base image, and let A and B be two additional foreground pieces joined to I. Let $\Delta E_A \coloneqq E(I \cup A) - E(I)$ and $\Delta E_B \coloneqq E(I \cup B) - E(I)$. Then:*

$$\Delta E_{A \cup B} - \Delta E_A - \Delta E_B = E(I \cap A \cap B) - E(A \cap B). \tag{12}$$

In particular, if $A \cap B = \emptyset$ (i.e., A and B are spatially disjoint), the joint topological change is strictly linear:

$$\Delta E_{A \cup B} = \Delta E_A + \Delta E_B. \tag{13}$$

Proof. Applying (8) to $(I \cup A) \cup (I \cup B) = I \cup A \cup B$:

$$\begin{aligned} E(I \cup A \cup B) &= E(I \cup A) + E(I \cup B) - E((I \cup A) \cap (I \cup B)) \\ &= E(I \cup A) + E(I \cup B) - E(I \cup (A \cap B)). \end{aligned}$$

Subtracting $E(I)$ from both sides and expanding $E(I \cup (A \cap B)) = E(I) + E(A \cap B) - E(I \cap A \cap B)$ completes the proof. □

2.4 Morse–Bott Critical Point Sweeps

Gray's critical point method can be formalized through continuous Morse theory [4, 5].

Definition 2.9 (Sweep Line and Linear Height Function)**.** Let $\theta \in S^1$ be a unit vector defining a sweep direction in $\mathbb{R}^2$. Define the linear height function $h_\theta : \mathbb{R}^2 \to \mathbb{R}$ by:

$$h_\theta(x) \coloneqq \langle x, \theta \rangle. \tag{14}$$

For any level $t \in \mathbb{R}$, let $H_t \coloneqq \{x \in \mathbb{R}^2 \mid h_\theta(x) = t\}$ denote the sweep line, and let $X_t \coloneqq X \cap \{x \in \mathbb{R}^2 \mid h_\theta(x) \le t\}$ denote the sub-level foreground.

Theorem 2.10 (Morse Index Theorem for Planar Binary Images)**.** *Assume ∂X is a smooth 1-manifold whose height function $h_\theta|_{\partial X}$ has only non-degenerate Morse critical points. As t increases from $-\infty$ to $+\infty$, the Euler characteristic $\chi(X_t)$ changes exclusively at critical points:*

1. ***Upstream Convexity** (X_θ): A local minimum of $h_\theta|_{\partial X}$ with outward normal vector pointing opposite to θ. Passing through X_θ attaches a 0-cell, creating a new connected component: $\Delta\chi = +1$.*
2. ***Upstream Concavity** (V_θ): A saddle point/local minimum of the complementary domain where the sweep line bridges two components or encloses a hole: $\Delta\chi = -1$.*
3. ***Downstream Extremities** ($X_{-\theta}, V_{-\theta}$): Local maxima of $h_\theta|_{\partial X}$ where boundary curves close. These satisfy $\Delta\chi = 0$.*

Consequently, summing all critical points across the image yields:

$$\chi(X) = B - L = n(X_\theta) - n(V_\theta). \tag{15}$$

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3 Planar Regular Tessellations and the Geometry of Cell Groups

3.1 Classification of Regular Planar Tessellations (p, q)

Definition 3.1 (Regular Planar Tessellation (p, q))**.** A *regular tessellation* of the Euclidean plane $\mathbb{R}^2$, denoted by the Schläfli symbol (p, q), is a cellular decomposition into regular convex p-sided polygons (**bit cells**) such that exactly q polygons meet at each vertex.

Proposition 3.2 (Schläfli Classification of Euclidean Lattices)**.** *The only regular tessellations of the Euclidean plane are:*

$$(6, 3) \textit{ (Hexagonal)}, \quad (4, 4) \textit{ (Square)}, \quad (3, 6) \textit{ (Triangular)}. \tag{16}$$

Proof. In a regular Euclidean p-gon, the interior vertex angle is $\alpha_p = \frac{(p-2)\pi}{p}$. For q such polygons to surround a vertex without gaps or overlaps, the angles must sum to 2π:

$$q \cdot \frac{(p-2)\pi}{p} = 2\pi \iff (p-2)(q-2) = 4. \tag{17}$$

For integers $p, q \geq 3$, the only positive integer factorizations of 4 are 1×4, 2×2, and 4×1, yielding $(p, q) \in \{(6, 3), (4, 4), (3, 6)\}$. □

3.2 Formal Definition of Dual Cell Groups

Definition 3.3 (Primal Vertices and Dual Cell Groups)**.** Let $\mathcal{T} = (p, q)$ be a regular planar tessellation.

- Let $\mathcal{C}_2(\mathcal{T})$ denote the set of open 2-cells (primal bit cells).
- Let $\mathcal{V}(\mathcal{T}) = \mathcal{C}_0(\mathcal{T})$ denote the set of primal 0-cells (grid vertices).
- For each primal vertex $v \in \mathcal{V}(\mathcal{T})$, let $\mathrm{Star}_2(v) = \{\sigma_2^{(1)}, \sigma_2^{(2)}, \ldots, \sigma_2^{(q)}\}$ be the set of q bit cells incident to v. The **cell group** $G(v)$ is the dual convex q-gon formed by connecting the geometric centers of these q bit cells in cyclic order:

$$G(v) \coloneqq \mathrm{conv}\left(\mathrm{center}(\sigma_2^{(1)}), \mathrm{center}(\sigma_2^{(2)}), \ldots, \mathrm{center}(\sigma_2^{(q)})\right). \tag{18}$$

Proposition 3.4 (Dual Voronoi Property of Cell Groups)**.** *The collection of cell groups $\{G(v)\}_{v \in \mathcal{V}(\mathcal{T})}$ forms the dual regular tessellation (q, p) of the plane. Each cell group $G(v)$ is geometrically centered at the primal vertex v.*

3.3 Linear Separability of Cell Groups and Connectivity Ambiguities

When a continuous binary image with smooth border curves is quantized onto a lattice (p, q), each bit cell takes state 1 or 0 depending on whether its center lies in the foreground.

Definition 3.5 (Linear Separability of Cell Groups)**.** A cell group configuration $(b_1, \ldots, b_q) \in \{0, 1\}^q$ is *linearly separable* if there exists an affine hyper-plane (a straight line in $\mathbb{R}^2$) separating all vertices with $b_i = 1$ from all vertices with $b_i = 0$. An image is *separable* if every cell group is linearly separable.

Theorem 3.6 (Separability Comparison Across Planar Lattices)**.** *1.* ***Hexagonal Lattice*** $(6, 3)$*: Cell groups are equilateral triangles ($q = 3$). In $\mathbb{R}^2$,* any *partition of the 3 vertices of a triangle into $\{0\}$ and $\{1\}$ is linearly separable. Thus, the inseparability count is* ***0 out of 8*** *(0%). Diagonal connectivity ambiguities are geometrically impossible.*

2. ***Square Lattice*** $(4,4)$*: Cell groups are squares* ($q=4$)*. Exactly 2 of the 16 binary patterns* (12.5%)*—namely the diagonally opposite configurations* $\begin{bmatrix}1&0\\0&1\end{bmatrix}$ *and* $\begin{bmatrix}0&1\\1&0\end{bmatrix}$ *(denoted* Q_D*)—are non-linearly separable (the XOR pattern).*

3. ***Triangular Lattice*** $(3,6)$*: Cell groups are regular hexagons* ($q=6$)*. Exactly 32 of the 64 binary patterns* (50%) *are non-linearly separable.*

Proof. For $q=3$, let $S \subset \{v_1, v_2, v_3\}$ be the set of 1-vertices. If $|S| \in \{1,2\}$, $\text{conv}(S)$ is either a point or a line segment, and $\text{conv}(\{v_1, v_2, v_3\} \setminus S)$ is a segment or a point disjoint from $\text{conv}(S)$. By the Hyperplane Separation Theorem, they are strictly linearly separable.

For $q=4$, the diagonals of a convex quad intersect at its interior center point $(1/2, 1/2)$. For the diagonal patterns Q_D, $\text{conv}(v_1, v_3) \cap \text{conv}(v_2, v_4) = \{(1/2, 1/2)\} \neq \emptyset$, precluding linear separation.

For $q=6$, any alternating binary pattern $(1,0,1,0,1,0)$ or pattern with 3 disjoint 1-vertices has intersecting convex hulls, resulting in 32 inseparable configurations out of $2^6 = 64$. □

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4 Discrete Local Valuations on Two-Dimensional Regular Lattices

4.1 The 2D Dual Decomposition Valuation Principle

Let $\mathcal{T} = (p,q)$ be a regular planar tessellation. To compute the global Euler characteristic $\chi(K(I)) = n_0 - n_1 + n_2$, we decompose the counts across the dual cell groups $\{G(v)\}_{v \in \mathcal{V}(\mathcal{T})}$.

Definition 4.1 (Vertex Multiplicity of k-Cells)**.** For a regular planar tessellation (p,q):

- $V_0 = 1$: Each 0-cell (vertex) is centered in exactly 1 dual cell group.
- $V_1 = 2$: Each 1-cell (unit edge) is bounded by 2 vertices, hence intersects exactly $V_1 = 2$ dual cell groups.
- $V_2 = p$: Each 2-cell (bit cell polygon) is bounded by p vertices, hence intersects exactly $V_2 = p$ dual cell groups.

Theorem 4.2 (2D Local Valuation Theorem)**.** *The global Euler characteristic, area, and perimeter of a digitized binary foreground* $K(I) \subset \mathcal{T}$ *are given by summing local cell-group valuations:*

$$\chi(K(I)) = \sum_{v \in \mathcal{V}(\mathcal{T})} \left(\Delta n_0(Q(v)) - \Delta n_1(Q(v)) + \Delta n_2(Q(v))\right), \tag{19}$$

$$A(K(I)) = \sum_{v \in \mathcal{V}(\mathcal{T})} \Delta n_2(Q(v)) \cdot \text{Area}(\sigma_2), \tag{20}$$

$$P(K(I)) = \sum_{v \in \mathcal{V}(\mathcal{T})} \Delta n_1^{\partial}(Q(v)) \cdot \text{Length}(e), \tag{21}$$

where $\Delta n_0(Q(v)) = \mathbf{1}_{\{Q(v) \neq \mathbf{0}\}}$, $\Delta n_1(Q(v)) = \frac{1}{2} n_1^{loc}(Q(v))$, $\Delta n_2(Q(v)) = \frac{1}{p} n_2^{loc}(Q(v))$, *and* Δn_1^{∂} *counts boundary edges.*

4.2 The Hexagonal Lattice $(6,3)$

On the hexagonal lattice Λ_{hex}, bit cells are regular hexagons ($p=6$) and cell groups are equilateral triangles ($q=3$).

4.2.1 Multiplicity Parameters

$$V_0 = 1, \quad V_1 = 2, \quad V_2 = 6. \tag{22}$$

Theorem 4.3 (Directional Critical Point Formula for Hexagons)**.** *Choosing a sweep direction θ parallel to a hexagon edge orientation:*

$$n(X_\theta) = n\begin{bmatrix} & 1 & \\ 0 & & 0 \end{bmatrix}, \qquad n(V_\theta) = n\begin{bmatrix} & 0 & \\ 1 & & 1 \end{bmatrix}. \tag{23}$$

By Theorem 2.10, the discrete Euler characteristic is:

$$E = n\begin{bmatrix} & 1 & \\ 0 & & 0 \end{bmatrix} - n\begin{bmatrix} & 0 & \\ 1 & & 1 \end{bmatrix}. \tag{24}$$

Theorem 4.4 (Isotropic Symmetric Hexagonal Triangle Formula)**.** *Let T_1 denote the total number of cell group triangles with exactly one* 1 *and two* 0*s, and let T_2 denote the total number of cell group triangles with exactly two* 1*s and one* 0*. Then:*

$$E = \frac{T_1 - T_2}{6}. \tag{25}$$

Proof. Let $Q(v) = (b_1, b_2, b_3) \in \{0,1\}^3$ denote the binary state of the cell group triangle $G(v)$. By Theorem 4.2:

- $Q = (0,0,0)$ (T_0 type): $\Delta n_0 = 0, \Delta n_1 = 0, \Delta n_2 = 0 \implies \Delta\chi = 0$.
- Q has one 1 (T_1 type): $\Delta n_0 = 1$, $\Delta n_1 = 2 \times \frac{1}{2} = 1$, $\Delta n_2 = 1 \times \frac{1}{6} = \frac{1}{6}$. Thus:

$$\Delta\chi(T_1) = 1 - 1 + \frac{1}{6} = +\frac{1}{6}.$$

- Q has two 1s (T_2 type): $\Delta n_0 = 1$, $\Delta n_1 = 3 \times \frac{1}{2} = \frac{3}{2}$, $\Delta n_2 = 2 \times \frac{1}{6} = \frac{1}{3}$. Thus:

$$\Delta\chi(T_2) = 1 - \frac{3}{2} + \frac{1}{3} = -\frac{1}{6}.$$

- $Q = (1,1,1)$ (T_3 type): $\Delta n_0 = 1$, $\Delta n_1 = 3 \times \frac{1}{2} = \frac{3}{2}$, $\Delta n_2 = 3 \times \frac{1}{6} = \frac{1}{2}$. Thus:

$$\Delta\chi(T_3) = 1 - \frac{3}{2} + \frac{1}{2} = 0.$$

Summing over all grid vertices $v \in \mathcal{V}(\mathcal{T})$ yields $E = \frac{1}{6}T_1 - \frac{1}{6}T_2 = \frac{T_1 - T_2}{6}$. □

Theorem 4.5 (Hexagonal Perimeter and Cauchy–Crofton Approximation)**.** *The exact digital boundary perimeter P is:*

$$P = 2n[0\ 1] + 2n\begin{bmatrix} 0 \\ 1 \end{bmatrix} + 2n\begin{bmatrix} 0 & \\ & 1 \end{bmatrix} = T_1 + T_2. \tag{26}$$

The orientation-averaged continuous perimeter P_a satisfying zero expected error for smooth curves is:

$$P_a = \frac{T_1 + T_2}{1.24} \approx \frac{3}{\pi}(T_1 + T_2). \tag{27}$$

4.3 The Square Lattice $(4,4)$

On the square grid $\mathbb{Z}^2$, bit cells are unit squares ($p = 4$) and cell groups are squares ($q = 4$) connecting the 4 centers of bit cells meeting at grid intersection v.

4.3.1 Multiplicity Parameters

$$V_0 = 1, \quad V_1 = 2, \quad V_2 = 4. \tag{28}$$

A 2×2 bit quad contains $2^4 = 16$ configurations, partitioned into 6 rotation equivalence classes:

$$Q_0 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad Q_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad Q_2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \tag{29}$$

$$Q_D = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}, \qquad Q_3 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \qquad Q_4 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}. \tag{30}$$

Table 1: Fractional cell allocations and local Euler valuations for square lattice bit quads.

Quad Class	**Multiplicity**	Δn_0	Δn_1	Δn_2	$\Delta\chi_W$ (8-conn)	$\Delta\chi_Z$ (4-conn)
Q_0	1	0	0	0	0	0
Q_1	4	1	$2 \times \frac{1}{2} = 1$	$1 \times \frac{1}{4} = \frac{1}{4}$	$1 - 1 + \frac{1}{4} = +\frac{1}{4}$	$+\frac{1}{4}$
Q_2	4	1	$3 \times \frac{1}{2} = \frac{3}{2}$	$2 \times \frac{1}{4} = \frac{1}{2}$	$1 - \frac{3}{2} + \frac{1}{2} = 0$	0
Q_D (8-conn)	2	1	$4 \times \frac{1}{2} = 2$	$2 \times \frac{1}{4} = \frac{1}{2}$	$1 - 2 + \frac{1}{2} = -\frac{1}{2}$	–
Q_D (4-conn)	2	2	$4 \times \frac{1}{2} = 2$	$2 \times \frac{1}{4} = \frac{1}{2}$	–	$2 - 2 + \frac{1}{2} = +\frac{1}{2}$
Q_3	4	1	$3 \times \frac{1}{2} = \frac{3}{2}$	$3 \times \frac{1}{4} = \frac{3}{4}$	$1 - \frac{3}{2} + \frac{3}{4} = -\frac{1}{4}$	$-\frac{1}{4}$
Q_4	1	1	$2 \times \frac{1}{2} = 1$	$4 \times \frac{1}{4} = 1$	$1 - 1 + 1 = 0$	0

Theorem 4.6 (Square Lattice Euler Characteristic Formulas)**.** *Summing the fractional valuations over all quads in the image:*

$$4W = n(Q_1) - n(Q_3) - 2n(Q_D) \quad \textit{(8-connectivity / 1s-joined)}, \tag{31}$$

$$4Z = n(Q_1) - n(Q_3) + 2n(Q_D) \quad \textit{(4-connectivity / 0s-joined)}, \tag{32}$$

$$4H = n(Q_1) - n(Q_3) \quad \textit{(Symmetric half-connected model).} \tag{33}$$

Proof. By Table 1:

$$W = \sum_{v \in \mathbb{Z}^2} \Delta\chi_W(Q(v)) = \frac{1}{4}n(Q_1) + 0n(Q_2) - \frac{1}{2}n(Q_D) - \frac{1}{4}n(Q_3) + 0n(Q_4).$$

Multiplying by 4 yields (31). Under 4-connectivity, the two diagonal 1s do not share a vertex in the cellular complex, splitting v into two vertices ($\Delta n_0 = 2$), yielding $\Delta\chi_Z(Q_D) = 2 - 2 + 1/2 = +1/2$. Multiplying by 4 gives (32). The average $H = \frac{W+Z}{2}$ eliminates the $n(Q_D)$ term, yielding (33). □

Theorem 4.7 (Area and Perimeter Valuations on Square Grids)**.**

$$4A = n(Q_1) + 2n(Q_2) + 2n(Q_D) + 3n(Q_3) + 4n(Q_4) = 4n([1]), \tag{34}$$

$$P = n(Q_1) + n(Q_2) + 2n(Q_D) + n(Q_3) = 2n[0\ 1] + 2n \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{35}$$

For smooth continuous boundaries, corner-cutting approximations yield:

$$P_a = n(Q_2) + \frac{n(Q_1) + n(Q_3)}{\sqrt{2}}, \tag{36}$$

$$4A_a = \frac{1}{2}n(Q_1) + n(Q_2) + n(Q_D) + \frac{7}{2}n(Q_3) + 4n(Q_4). \tag{37}$$

4.4 The Triangular Lattice $(3, 6)$

On the triangular lattice $(3, 6)$, bit cells are triangles ($p = 3$) and cell groups are regular hexagons ($q = 6$).

- Multiplicity parameters: $V_0 = 1, V_1 = 2, V_2 = 3$.
- Each cell group hexagon contains vertex v ($\Delta n_0 = 1$), $\frac{1}{2}$ of each incident edge, and $\frac{1}{3}$ of each incident triangle.
- The local contribution formula is $\Delta\chi(Q(v)) = \Delta n_0 - \frac{1}{2}n_1^{\text{loc}} + \frac{1}{3}n_2^{\text{loc}}$.

—

5 Local Differentials and Parallel Skeletonization

5.1 Definition and Locality of Differentials

Definition 5.1 (Euler, Area, and Perimeter Differentials)**.** Let U be a bit cell in an image I. The *Euler differential* $E^*(U)$, *area differential* $A^*(U)$, and *perimeter differential* $P^*(U)$ are:

$$E^*(U) \coloneqq \frac{\Delta E}{\Delta U}, \quad A^*(U) \coloneqq \frac{\Delta A}{\Delta U}, \quad P^*(U) \coloneqq \frac{\Delta P}{\Delta U}, \tag{38}$$

where $\Delta U = +1$ when U changes from $0 \to 1$, and $\Delta U = -1$ when U changes from $1 \to 0$.

Theorem 5.2 (Locality of Differentials)**.** *For a regular tessellation (p, q), changing bit cell U modifies exactly the p cell groups centered at the p boundary vertices of U. Consequently, $E^*(U), A^*(U)$, and $P^*(U)$ are strictly functions of the immediate neighborhood ring surrounding U.*

5.2 Hexagonal Differentials

Let $\mathcal{N}_6(U) = (U_0, U_1, \ldots, U_5)$ be the 6 cyclic neighbors of U.

Theorem 5.3 (Hexagonal Differential Theorem)**.** *Let $n(0, 1)$ be the number of $0 \to 1$ transitions in cyclic order $(U_0, \ldots, U_5, U_0)$. Then:*

$$E^*(U) = 1 - n(0, 1) = 1 - \sum_{i=0}^{5} \bar{U}_i U_{i+1} = 1 - |U'|, \tag{39}$$

where $U'_i \coloneqq \bar{U}_i \wedge U_{i+1}$ (with indices modulo 6). Furthermore:

$$A^*(U) = 1, \qquad P^*(U) = 6 - 2|U|, \tag{40}$$

where $|U| = \sum_{i=0}^{5} U_i$ is the neighbor population.

Proof. When $U : 0 \to 1$:

- Exactly one 2-cell is added: $\Delta n_2 = +1$.
- A boundary unit edge is added for every 0-neighbor: $\Delta n_1 = n(0)$.
- A boundary vertex is added for every consecutive pair of zero neighbors: $\Delta n_0 = n(0, 0)$.

Thus $\Delta E = n(0, 0) - n(0) + 1$. Using the cyclic identity $n(0) = n(0, 0) + n(0, 1)$, we obtain $E^*(U) = 1 - n(0, 1)$. For perimeter, adding U creates 6 potential edges and removes 2 edges for each active neighbor, giving $P^*(U) = 6 - 2|U|$. □

Proposition 5.4 (Topological Taxonomy of $E^*(U)$)**.**

- $E^*(U) = +1$*: U creates an isolated component (if $|U| = 0$) or eliminates a hole (if $|U| = 6$).*
- $E^*(U) = 0$*: U is a* ***simple point****. Setting or clearing U preserves all topological Betti numbers.*
- $E^*(U) = -1$*: U joins two disconnected components or closes a loop (creating a hole).*
- $E^*(U) \leq -2$*: U is a branch point joining ≥ 3 components.*

5.3 Square Lattice Differentials

For the 8-neighborhood $\mathcal{N}_8(U) = (U_0, \ldots, U_7)$ on $\mathbb{Z}^2$ (with even indices for edge neighbors and odd indices for corner neighbors):

Theorem 5.5 (Square Lattice Differential Formulas)**.** *Let C_{246} count corner triplets matching $\{0,1,0\}, \{1,0,0\}, \{1,1,0\}$. Let C_{135} count corner triplets matching $\{0,0,1\}, \{0,1,1\}, \{1,0,1\}$. Let C_{07} count homogeneous corners ($\{0,0,0\}$ or $\{1,1,1\}$). Then:*

$$W^*(U) = 1 - C_{246} = 1 - |U'_e|, \tag{41}$$

$$Z^*(U) = 1 - C_{135} = 1 - |U'_d|, \tag{42}$$

$$H^*(U) = \frac{1}{2}C_{07} - 1 = 1 - \frac{1}{2}|U'|, \tag{43}$$

$$P^*(U) = 4 - 2|U_d|, \tag{44}$$

where $|U_d| = \sum_{k=0}^{3} U_{2k}$ is the 4-neighbor population.

5.4 Subfield Decomposition and Parallel Skeletonization

Theorem 5.6 (Subfield Thinning Theorem)**.** *Let Λ be a regular lattice partitioned into M disjoint sublattices $\Lambda = \bigcup_{k=1}^{M} \Lambda^{(k)}$ such that no two cells in any single sublattice $\Lambda^{(k)}$ share a cell group. Then simultaneously removing all simple points ($E^*(U) = 0$) within $\Lambda^{(k)}$ satisfies:*

$$\Delta E_{batch} = \sum_{U \in \Lambda^{(k)}, E^*(U)=0} E^*(U) = 0. \tag{45}$$

The minimal number of subfields required for collision-free parallel thinning is:

$$M_{hex} = 3, \quad M_{square} = 4, \quad M_{tri} = 6, \quad M_{\mathbb{Z}^N} = 2^N. \tag{46}$$

Proof. By Theorem 2.8, if all modified cells in a batch are mutually disjoint ($A \cap B = \emptyset$), the joint valuation change is strictly linear. Partitioning the lattice into independent cosets ensures all simultaneous updates are disjoint. □

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6 Geometric Shape Characterization via Locally Countable Properties

By combining the locally countable functionals E, A, and P, one obtains powerful scale- and affine-invariant shape descriptors.

Definition 6.1 (Component-Averaged Metrics)**.** Assuming the number of holes is small relative to objects ($E \approx B$):

$$\bar{A} \coloneqq \frac{A}{E} \quad \text{(Average Area per Object)}, \qquad \bar{P} \coloneqq \frac{P}{E} \quad \text{(Average Perimeter per Object)}. \tag{47}$$

Theorem 6.2 (Geometric Descriptors for Fibrous and Elongated Structures)**.** *For long, thin, line-like structures (e.g., characters, fibers, vascular networks), the continuous line length k, average line width w, and slenderness aspect ratio k/w are:*

$$k = \frac{P}{2}, \qquad w = \frac{A}{k} = \frac{2A}{P}, \qquad \frac{k}{w} = \frac{P^2}{4A}. \tag{48}$$

Theorem 6.3 (Isoperimetric Circularity and Dispersion)**.** *By the Classical Isoperimetric Inequality in $\mathbb{R}^2$, for any plane region with area A and perimeter P:*

$$P^2 \geq 4\pi A \iff \mathcal{D} \coloneqq \frac{P^2}{4\pi A} \geq 1, \tag{49}$$

with equality $\mathcal{D} = 1$ achieved if and only if the object is a Euclidean circular disc.

Remark 6.4 (OCR Quality Acceptance Criteria). As established by Bartz (1968) [11] and Gray (1971) [1], high-quality optical character recognition requires:

$$\frac{w}{W_{\text{char}}} \in [0.10, 0.20], \quad \frac{k}{w} \geq 10, \quad E \in \{-1, 0, 1, 2\}. \tag{50}$$

—

7 N-Dimensional Dual Voronoi Valuation Framework

We now generalize the cell-group valuation framework to arbitrary N-dimensional uniform and regular cellular complexes.

7.1 Geometric and Topological Setup

Let $\mathbb{M}^N$ be an N-dimensional simply connected Riemannian manifold of constant curvature ($\mathbb{R}^N$, $\mathbb{S}^N$, or $\mathbb{H}^N$).

Definition 7.1 (N-Dimensional Regular/Uniform Cellular Complex)**.** Let $\mathcal{T}$ be a locally finite, regular or uniform cellular decomposition of $\mathbb{M}^N$. For each dimension $k \in \{0, 1, \ldots, N\}$, let $\mathcal{C}_k(\mathcal{T})$ denote the set of open k-cells. The top-dimensional cells $\sigma_N \in \mathcal{C}_N(\mathcal{T})$ are termed *generalized voxels*.

Definition 7.2 (N-Dimensional Binary Image and Foreground Complex)**.** A *binary image* on $\mathcal{T}$ is a compactly supported map $I : \mathcal{C}_N(\mathcal{T}) \to \{0, 1\}$. The *foreground complex* $K(I) \subseteq \mathcal{T}$ is the closed subcomplex:

$$K(I) \coloneqq \bigcup_{\substack{\sigma_N \in \mathcal{C}_N(\mathcal{T}) \\ I(\sigma_N)=1}} \bar{\sigma}_N. \tag{51}$$

The set of active k-cells is $\mathcal{C}_k(K(I)) \coloneqq \{\sigma_k \in \mathcal{C}_k(\mathcal{T}) \mid \sigma_k \subseteq K(I)\}$, and $n_k(K(I)) \coloneqq |\mathcal{C}_k(K(I))|$.

Definition 7.3 (Euler–Poincaré Characteristic)**.**

$$\chi(K(I)) = \sum_{k=0}^{N} (-1)^k n_k(K(I)). \tag{52}$$

7.2 The Dual Voronoi Partition

Let $\mathcal{V}(\mathcal{T}) = \mathcal{C}_0(\mathcal{T})$ denote the primal 0-cells of $\mathcal{T}$.

Definition 7.4 (Dual Voronoi Cell Group)**.** For each primal vertex $v \in \mathcal{V}(\mathcal{T})$, the *dual Voronoi cell* $C(v)$ is:

$$C(v) \coloneqq \left\{ x \in \mathbb{M}^N \;\middle|\; d(x,v) \leq d(x,w) \quad \forall w \in \mathcal{V}(\mathcal{T}) \right\}. \tag{53}$$

Definition 7.5 (Vertex Multiplicity of k-Cells)**.** For a regular or uniform tessellation $\mathcal{T}$, each k-cell $\sigma_k \in \mathcal{C}_k(\mathcal{T})$ is spanned by a fixed number of boundary vertices:

$$V_k \coloneqq |\mathcal{V}(\bar{\sigma}_k)| = |\mathcal{C}_0(\bar{\sigma}_k)|. \tag{54}$$

Lemma 7.6 (Uniform Multiplicity of Cell Partitioning)**.** *Let $\{C(v)\}_{v \in \mathcal{V}(\mathcal{T})}$ be the dual Voronoi partition of $\mathbb{M}^N$. For every $k \in \{0, \ldots, N\}$ and every k-cell $\sigma_k \in \mathcal{C}_k(\mathcal{T})$:*

$$\sum_{v \in \mathcal{V}(\mathcal{T})} \mathbf{1}_{\{\sigma_k \cap \operatorname{int}(C(v)) \neq \emptyset\}} = V_k. \tag{55}$$

Proof. By vertex-transitivity and symmetry of $\mathcal{T}$:

1. For $k = 0$, v is the center of $C(v)$, so $V_0 = 1$.
2. For $k = 1$, any edge $e = (v_1, v_2)$ is bisected by $\partial C(v_1) \cap \partial C(v_2)$, intersecting exactly $V_1 = 2$ dual cells.
3. For $k \geq 2$, the barycentric subdivision of σ_k divides the interior into V_k congruent sectors, each contained in the Voronoi cell $C(v_i)$ of one of its V_k spanning vertices $v_i \in \mathcal{V}(\bar{\sigma}_k)$. □

7.3 The Universal N-Dimensional Local Valuation Theorem

Definition 7.7 (Local Neighborhood Configuration $Q(v)$)**.** For each primal vertex $v \in \mathcal{V}(\mathcal{T})$, let $\operatorname{Star}_N(v) = \{\sigma_N^{(1)}, \ldots, \sigma_N^{(D_v)}\}$ be the set of D_v distinct N-cells meeting at v. The local configuration vector is:

$$Q(v) \coloneqq \left(I(\sigma_N^{(1)}), I(\sigma_N^{(2)}), \ldots, I(\sigma_N^{(D_v)}) \right) \in \{0,1\}^{D_v}. \tag{56}$$

Theorem 7.8 (Universal N-Dimensional Local Valuation Theorem)**.** *The global Euler characteristic $\chi(K(I))$ is identically equal to the sum of local Euler valuations evaluated over all primal vertices:*

$$\chi(K(I)) = \sum_{v \in \mathcal{V}(\mathcal{T})} \Delta\chi(Q(v)), \tag{57}$$

where:

$$\Delta\chi(Q(v)) = \sum_{k=0}^{N} (-1)^k \Delta n_k(Q(v)), \quad \textit{with} \quad \Delta n_k(Q(v)) \coloneqq \frac{1}{V_k} \cdot \left| \{\sigma_k \in \mathcal{C}_k(K(I)) \mid v \in \bar{\sigma}_k\} \right|. \tag{58}$$

Proof. Summing over all vertices $v \in \mathcal{V}(\mathcal{T})$ and swapping the summation order:

$$\begin{aligned}
\sum_{v \in \mathcal{V}(\mathcal{T})} \Delta\chi(Q(v)) &= \sum_{v \in \mathcal{V}(\mathcal{T})} \sum_{k=0}^{N} (-1)^k \Delta n_k(Q(v)) \\
&= \sum_{k=0}^{N} (-1)^k \sum_{v \in \mathcal{V}(\mathcal{T})} \frac{1}{V_k} \sum_{\substack{\sigma_k \in \mathcal{C}_k(K(I)) \\ v \in \bar{\sigma}_k}} 1 \\
&= \sum_{k=0}^{N} (-1)^k \frac{1}{V_k} \sum_{\sigma_k \in \mathcal{C}_k(K(I))} \sum_{v \in \mathcal{V}(\bar{\sigma}_k)} 1.
\end{aligned}$$

By definition of V_k, the innermost sum is $\sum_{v\in\mathcal{V}(\bar{\sigma}_k)} 1 = V_k$. Therefore:

$$\sum_{v\in\mathcal{V}(\mathcal{T})} \Delta\chi(Q(v)) = \sum_{k=0}^{N}(-1)^k \frac{1}{V_k} \sum_{\sigma_k\in\mathcal{C}_k(K(I))} V_k = \sum_{k=0}^{N}(-1)^k n_k(K(I)) = \chi(K(I)). \qquad \square$$

Theorem 7.9 (Forman's Discrete Morse Spherical Link Formula)**.** *Let $S_\varepsilon^{N-1}(v)$ be a geodesic sphere of radius $\varepsilon > 0$ centered at $v \in \mathcal{V}(\mathcal{T})$, and let $\mathrm{Lk}(v, K(I)) \coloneqq K(I) \cap S_\varepsilon^{N-1}(v)$ denote the spherical link. Then:*

$$\Delta\chi(Q(v)) = \begin{cases} 1 - \chi(\mathrm{Lk}(v, K(I))) & \text{if } Q(v) \neq \mathbf{0}, \\ 0 & \text{if } Q(v) = \mathbf{0}. \end{cases} \tag{59}$$

—

8 Specialization to 3-Dimensional Volume Processing on $\mathbb{Z}^3$

Let $\mathcal{T} = \mathbb{Z}^3$ be the standard 3D cubic lattice. Each voxel is a unit cube $\sigma_3 = [x, x+1] \times [y, y+1] \times [z, z+1]$.

8.1 Combinatorial Parameters and Denominator Clearing

- $V_0 = 1$ (1 vertex per 0-cell).
- $V_1 = 2$ (2 vertices per unit edge).
- $V_2 = 4$ (4 vertices per square face).
- $V_3 = 8$ (8 vertices per cube voxel).
- At every grid vertex $v \in \mathbb{Z}^3$, exactly $D_v = 2^3 = 8$ voxels meet.

The local configuration $Q(v) \in \{0,1\}^{2\times2\times2}$ is a $2\times2\times2$ **bit octocube** ($2^8 = 256$ local binary states).

Corollary 8.1 (Universal Hypercubic Normalization)**.** *Multiplying by* $\mathrm{lcm}(V_0, V_1, V_2, V_3) = 8$ *clears all fractional denominators, yielding the exact integer formula:*

$$8\chi(K(I)) = \sum_{v\in\mathbb{Z}^3} \left(8\Delta n_0(Q(v)) - 4n_1^{loc}(Q(v)) + 2n_2^{loc}(Q(v)) - n_3^{loc}(Q(v))\right). \tag{60}$$

8.2 Complete Classification of the 22 Octocube Equivalence Classes

Under the full octahedral symmetry group O_h of the cube (order 48), the 256 bit octocube configurations partition into exactly **22 equivalence classes**.

8.3 3D Surface Area Valuation

Theorem 8.2 (3D Digital Boundary Surface Area)**.** *The continuous boundary surface area $S(K(I))$ is locally computable by counting boundary square faces:*

$$S(K(I)) = \sum_{v\in\mathbb{Z}^3} \Delta S(Q(v)), \quad \textit{with} \quad \Delta S(Q(v)) = \frac{1}{4} \cdot |\{\textit{active boundary 2-faces incident to } v\}|. \tag{61}$$

—

Table 2: Classification of all 256 octocube configurations under O_h symmetry into 22 equivalence classes, with local cell allocations and integer valuations $8\Delta\chi(Q)$.

Class ID	**Voxel Count** n_3	**Multiplicity**	n_2^{loc}	n_1^{loc}	n_0^{loc} (26-conn)	$8\Delta\chi(Q)$
C_0	0	1	0	0	0	0
C_1	1	8	3	3	1	$8(1) - 4(3) + 2(3) - 1 = +1$
C_2	2 (face-adj)	12	5	5	1	$8(1) - 4(5) + 2(5) - 2 = 0$
C_3	2 (edge-adj)	12	6	5	1	$8(1) - 4(5) + 2(6) - 2 = +2$
C_4	2 (body-diag)	4	6	6	1	$8(1) - 4(6) + 2(6) - 2 = +3$
C_5	3 (corner)	24	7	6	1	$8(1) - 4(6) + 2(7) - 3 = +1$
C_6	3 (inline L)	24	8	7	1	$8(1) - 4(7) + 2(8) - 3 = +3$
C_7	3 (skew)	8	9	8	1	$8(1) - 4(8) + 2(9) - 3 = +5$
C_8	4 (square face)	6	8	6	1	$8(1) - 4(6) + 2(8) - 4 = 0$
C_9	4 (tripod)	24	9	7	1	$8(1) - 4(7) + 2(9) - 4 = +2$
C_{10}	4 (chair)	24	10	8	1	$8(1) - 4(8) + 2(10) - 4 = +4$
C_{11}	4 (skew ring)	12	10	8	1	$8(1) - 4(8) + 2(10) - 4 = +4$
C_{12}	4 (tetrahedral)	2	12	10	1	$8(1) - 4(10) + 2(12) - 4 = +8$
C_{13}	5 (compl C_5)	24	10	7	1	$8(1) - 4(7) + 2(10) - 5 = +1$
C_{14}	5 (compl C_6)	24	11	8	1	$8(1) - 4(8) + 2(11) - 5 = +3$
C_{15}	5 (compl C_7)	8	12	9	1	$8(1) - 4(9) + 2(12) - 5 = +5$
C_{16}	6 (compl C_2)	12	11	7	1	$8(1) - 4(7) + 2(11) - 6 = 0$
C_{17}	6 (compl C_3)	12	12	8	1	$8(1) - 4(8) + 2(12) - 6 = +2$
C_{18}	6 (compl C_4)	4	12	8	1	$8(1) - 4(8) + 2(12) - 6 = +2$
C_{19}	7 (compl C_1)	8	12	6	1	$8(1) - 4(6) + 2(12) - 7 = -1$
C_{20}	8 (full cube)	1	12	6	1	$8(1) - 4(6) + 2(12) - 8 = 0$
C_{21}	4 (diagonal pair)	12	10	9	1	$8(1) - 4(9) + 2(10) - 4 = +6$

9 Theory of Local Computability and Local Property Counters (LPC)

9.1 The Local Property Counter (LPC) Formalism

To formally characterize the complexity of local counting algorithms, Gray defined the **Local Property Counter** (LPC).

Definition 9.1 (LPC Complexity 4-Tuple)**.** A local measurement algorithm Φ on an N-dimensional lattice Λ is characterized by the tuple:

$$\mathcal{L}(\Phi) \coloneqq (D, G, R, M) \in \mathbb{Z}_+^4, \tag{62}$$

where:

- **Diameter (D)**: The maximum spatial coordinate span $\max(I_1, \ldots, I_N)$ of the local neighborhood window.
- **Degree (G)**: The number of distinct Boolean logic predicates $\{F_1, \ldots, F_G\}$ feeding the property accumulator.
- **Order (R)**: The maximum number of input bit cells feeding any single Boolean predicate: $R \coloneqq \max_g |\operatorname{supp}(F_g)|$.
- **Dimension (M)**: The maximum number of orthogonal coordinate axes spanned by any single predicate F_g.

9.2 Fundamental Complexity Theorems for Local Counting

Theorem 9.2 (Universal Diameter-2 Sufficiency)**.** *On any hypercubic lattice $\mathbb{Z}^N$, the topological Euler characteristic χ and all Hadwiger intrinsic valuations (A, P, S, V_N) are computable with*

diameter:

$$D \leq 2. \tag{63}$$

Proof. By Theorem 7.8, $\Phi(K(I)) = \sum_{v \in \mathbb{Z}^N} \Delta\Phi(Q(v))$. The local configuration $Q(v)$ consists of the 2^N hypercubes sharing primal vertex v, which spans exactly a $2 \times 2 \times \cdots \times 2$ window. The diameter is $\max(2, \ldots, 2) = 2$. □

Theorem 9.3 (Order Complexity Comparison)**.** *1. For **directional sweep / critical point formulas**, the Euler characteristic has order:*

$$R = N + 1. \tag{64}$$

*2. For **isotropic dual Voronoi cell valuations**, the Euler characteristic has order:*

$$R = 2^N. \tag{65}$$

Proof. A directional Morse sweep requires sampling N independent coordinate rays relative to a base point, which requires $N + 1$ points in general position. Fully symmetric isotropic valuations require sampling all 2^N hypercube quadrants meeting at v. □

9.3 Master Classification Table of Local Invariants Across Dimensions

Table 3: Comprehensive classification of local property counters across dimensions and lattices.

Property	**Lattice**	**Algorithm / Formula**	D	G	R	M
1D Length	$\mathbb{Z}^1$	$L = n[1]$	1	1	1	0
1D Euler	$\mathbb{Z}^1$	$E = n[01]$	2	1	2	1
2D Area	$(4,4),(6,3)$	$A = n[1]$	1	1	1	0
2D Approx Area	$(4,4)$	Corner-cutting A_a	2	5	4	2
2D Perimeter	$(6,3)$	$P = T_1 + T_2$	2	3	2	1
2D Perimeter	$(4,4)$	$P = 2n[01] + 2n[^0_1]$	2	2	2	1
2D Approx Perimeter	$(6,3)$	$P_a = \frac{T_1+T_2}{1.24}$	2	3	2	1
2D Approx Perimeter	$(4,4)$	$P_a = n(Q_2) + \frac{n(Q_1)+n(Q_3)}{\sqrt{2}}$	2	3	4	2
2D Euler (Critical Pt)	$(6,3)$	$E = n[\cdot] - n[\cdot]$	2	2	3	2
2D Euler (Critical Pt)	$(4,4)$	$E = n[\cdot] - n[\cdot]$	2	2	3	2
2D Euler (Isotropic)	$(6,3)$	$E = \frac{T_1-T_2}{6}$	2	2	3	2
2D Euler W (8-conn)	$(4,4)$	$4W = n(Q_1) - n(Q_3) - 2n(Q_D)$	2	3	4	2
2D Euler Z (4-conn)	$(4,4)$	$4Z = n(Q_1) - n(Q_3) + 2n(Q_D)$	2	3	4	2
2D Euler H (Half)	$(4,4)$	$4H = n(Q_1) - n(Q_3)$	2	2	4	2
2D Euler (Isotropic)	$(3,6)$	$\Delta\chi = \Delta n_0 - \frac{1}{2}n_1 + \frac{1}{3}n_2$	2	6	6	2
3D Volume	$\mathbb{Z}^3$	$V = n[1]$	1	1	1	0
3D Surface Area	$\mathbb{Z}^3$	$S = \sum \Delta S(Q(v))$	2	3	2	1
3D Euler (Critical Pt)	$\mathbb{Z}^3$	Morse sweep along 3 axes	2	4	4	3
3D Euler (Isotropic)	$\mathbb{Z}^3$	$8\chi = \sum \Delta\chi(Q(v))$ (Table 2)	2	22	8	3
ND Volume	$\mathbb{Z}^N$	$V_N = n[1]$	1	1	1	0
ND Euler (Critical Pt)	$\mathbb{Z}^N$	Morse sweep along N axes	2	$N+1$	$N+1$	N
ND Euler (Isotropic)	$\mathbb{Z}^N$	$2^N\chi = \sum \Delta\chi(Q(v))$	2	$\lvert\mathcal{O}(2^N)\rvert$	2^N	N

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10 Hardware Architecture of the Binary Image Processor (BIP)

Gray's theoretical results were directly implemented in the hardware of the **Binary Image Processor** (BIP) (Information International, Inc., 1970–1971) [1].

Definition 10.1 (BIP Raster Processing Architecture)**.** The BIP processes an image serially at 20 megabits/second by shifting a raster window $U_{i,j}$ sequentially across core memory. At each pixel (i, j), the hardware simultaneously evaluates:

1. A **result image** $R_{i,j} \coloneqq \mathcal{F}(U_{i,j}, M_{i,j}, P_{i,j})$, where $M_{i,j}$ is a template mask and $P_{i,j}$ is a programmable neighborhood majority predicate:

$$P_{i,j} \coloneqq C_{|U_{i,j} \wedge S|}, \tag{66}$$

   where $S = (S_0, \ldots, S_7) \in \{0, 1\}^8$ is an 8-bit selection register and $C = (C_0, \ldots, C_8) \in \{0, 1\}^9$ is a count register.

2. A bank of parallel **Local Property Counters** (LPC) incrementing property registers $\Phi \leftarrow \Phi + W_g$ whenever $F_g(U_{i,j}) = 1$.

Theorem 10.2 (Single-Cycle Hardware Evaluation of Differentials)**.** *By programming S to select the even subfield U_{even} and loading C with the lookup table for corner transitions C_{246}, the BIP computes the exact 1s-joined Euler differential $W^*(U)$ in a single hardware clock cycle, enabling real-time topological thinning and feature extraction.*

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11 Conclusion

This monograph has developed a unified, mathematically rigorous formalization of Stephen B. Gray's 1971 paper and extended it to general N-dimensional cellular complexes and hypercubic lattices $\mathbb{Z}^N$.

We have demonstrated:

1. The Euler characteristic χ, perimeter P, and area A are intrinsic Hadwiger valuations that decompose into local cell valuations across regular planar tessellations $(6, 3), (4, 4), (3, 6)$ and hypercubic lattices $\mathbb{Z}^N$ via dual Voronoi cell groups.

2. The linear separability of cell groups provides a geometric foundation for digital connectivity, explaining why the hexagonal lattice is universally unambiguous (0% inseparability), whereas the square lattice exhibits diagonal ambiguities (12.5% inseparability).

3. The Euler differential $E^*(U) = \Delta E / \Delta U$ locally classifies simple points, enabling topology-preserving parallel skeletonization via M-subfield decompositions ($M_{\text{hex}} = 3, M_{\text{square}} = 4, M_{\mathbb{Z}^N} = 2^N$).

4. In N dimensions, local valuations clear all fractional denominators via the universal factor 2^N, connecting directly to Forman's discrete Morse spherical link formula $\Delta\chi(Q) = 1 - \chi(\text{Lk}(v, K))$.

5. Local property counting is governed by universal diameter-2 sufficiency ($D \leq 2$), providing a sharp complexity characterization for local computability in discrete geometry.

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